Dialectal Bias in Hate Speech Detection Systems

Sérgio Apolinário da Costa

Department of Computing Science
Umeå University, Sweden
mai25sca@cs.umu.se

Abstract. This project audits two openly available classifiers, a hate speech detector and a toxicity detector, on three Twitter corpora: Davidson et al., the Twitter part of HateXplain and the US random sample of HateDay. Tweets are grouped by how closely their wording matches African-American English (AAE) with the dialect model of Blodgett et al. On tweets that humans labelled *neither hate nor offensive*, the AAE-aligned group is flagged more often by both classifiers in all three corpora (false positive rate gaps from +0.02 to +0.30, all intervals above zero), and a random AAE-aligned tweet gets a higher score than a random mainstream one with probability 0.62 to 0.75.

1 Introduction

Social media platforms rely on automatic classifiers to flag hateful content. This systems are usually trained and tested on text close to mainstream written English. Friedman and Nissenbaum call a system biased when it works well for the population it was built and tested on, but fails for users who differ from it. African-American English (AAE) is a rule-governed variety of English with its own grammar and vocabulary. Earlier work found that hate speech datasets, and the models trained on them, treat tweets associated with AAE differently. This project audits two openly available classifiers, on three different datasets.

1.1 Research Questions

This project will address the following research questions:

- Do the classifiers flag AAE-aligned tweets that are *not* hate more often than mainstream tweets?
- Does the gap appear for both classifiers, not only one?
- Does the gap appear in all three corpora?
- Does the gap stay when we use a different threshold?
- Does the gap stay when we define the groups in another way?
- Is the gap caused by the label mix of the two groups?
- Is the gap caused by profanity?
- Is the gap caused by tweet length?
- Do the words in the tweet explain the gap?
- Which words go with the false positives?

The hypothesis is that the classifiers flag more of the AAE-aligned tweets that are not hate than of the mainstream ones.

2 Literature Review

2.1 Automated Hate Speech Detection and the Problem of Offensive Language

Davidson et al. [2] collected tweets that match a hate speech lexicon and had crowd workers label them as *hate speech*, *offensive language* or *neither*. Their main point is that systems tend to confuse offensive language with hate speech. This distinction is central here. A classifier may disagree with the labels because its definition of hate differs, not because it is wrong. The main idea is that offensive language is not hate speech.

2.2 Demographic Dialectal Variation in Social Media: A Case Study of AAE

Blodgett et al. [1] built a mixed-membership model from geolocated tweets and Census demographics. For a tweet, it gives proportions over four demographically aligned language groups, one of which is African-American-aligned. The output describes how much a text resembles the usage of a group. It does not say who wrote it. In this project it is used as a measurement instrument to split tweets into AAE-aligned and mainstream groups, and this limit is stated wherever it matters.

2.3 The Risk of Racial Bias in Hate Speech Detection

Sap et al. [4] showed that AAE markers correlate with toxicity labels in hate speech datasets, that classifiers trained on these data propagate the correlation, and that annotators primed with dialect information label fewer tweets as offensive. Their results concern models *trained* on annotated corpora. In this project, it is used ready-made pretrained classifiers and it is evaluted on independent corpora. Following the usual reading of fairness as comparable error rates across groups, it was used the false positive rate (FPR) as the main quantity.

2.4 HateXplain: A Benchmark Dataset for Explainable Hate Speech Detection

HateXplain [3] is a benchmark of about 20,000 posts from Twitter and Gab. It has majority labels (hateful, offensive, normal), the targeted community and the words that justify each label. This project uses its Twitter part as a second corpus, because the dialect model was built on Twitter. It differs from Davidson in collection, annotation and text normalisation, so a gap found in both is less likely to be a property of one dataset.

2.5 HateDay: Insights from a Global Hate Speech Dataset Representative of a Day on Twitter

HateDay [5] is a random sample of all tweets posted on a single day (21 September 2022), annotated for hate speech. Unlike Davidson, which was selected with a lexicon, it reflects what a platform actually sees. Most tweets are harmless and hate is rare. This project only uses the US part as a third corpus to test whether the gap seen in lexicon-selected data also appears in everyday tweets. The dataset is gated, so no tweet is printed, quoted or committed in this project.

3 Materials and Methods

3.1 Overview

The pipeline, shown in Figure 1, has four different stages. Each stage is a group of scripts that writes files read by the next stage. The dialect model knows nothing about hate, the classifiers know nothing about dialect, and the human labels are used only at evaluation. In this project it was not trained any model. The code is available on: `github.com/serac01/hate-speech-detection-system`[1].

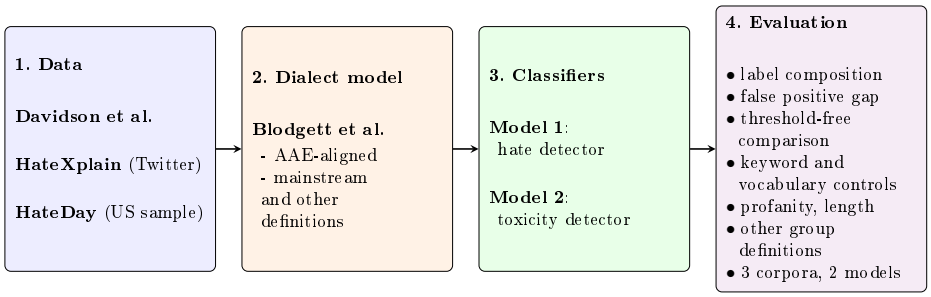


Fig. 1. Pipeline.

3.2 Data

All the three datasets share the label coding: *hate speech*, *offensive language*, *neither*. The Davidson dataset has 24,783 tweets (1,430 hate, 19,190 offensive, 4,163 neither), selected with a lexicon, so offensive vocabulary is over-represented. The Twitter extracted section from the HateXplain dataset has 8,777 posts with a majority label (normal = neither). Finnaly, the HateDay dataset with the US random sample has 26,970 tweets with a single annotation each. This dataset is gated on Hugging Face, so no tweet is printed, quoted or committed, and offensive words are masked in this report. The groups are estimated from the text and the corpora have no author information.

[1] `https://github.com/serac01/hate-speech-detection-system`

3.3 Dialect model

The Blodgett et al. [1] model was ported from Python 2 to 3 without changing inference. The AAE score from the model is bag-of-words, so it is not the probability that an author is African American. For this project it was setted a threshold for scoring, so a score ≥ 0.6 will be classified as *AAE-aligned* while a score ≤ 0.4 will be classified as *mainstream*. The rest is excluded. As robustness it was used the argmax of the four-way posterior (AAE vs. White), other thresholds, and a score computed after deleting profanity. This will be explored in more detail in the next section.

3.4 Classifiers

Both classifiers run through Hugging Face with a threshold of 0.5 and 128 tokens.

The model 1, `cardiffnlp/twitter-roberta-base-hate`[2], is a RoBERTa model pretrained on tweets and fine-tuned on the hate task of TweetEval, for example the HatEval (SemEval 2019 Task 5: hate against immigrants and women, tweets collected in 2018). Its positive class is *hate*, so it is compared with *hate speech*, the other two labels are negatives.

The model 2, `unitary/unbiased-toxic-roberta`[3] is another RoBERTa model fine-tuned on the Jigsaw "Unintended Bias in Toxicity Classification" on comments (online news comments, not tweets). It detects *toxicity*, not hate. So, for this project the positive class was mapped to *hate or offensive*. Its only real negatives are the tweets labelled as *neither*, and its flags on offensive tweets are correct detections. This model was chosen after a first candidate (a Dynabench hate model) was found to have been trained on data that includes Davidson's tweets, which would contaminate the test, so it was replaced.

Both datasets are independence of the test data. The model 2 was trained on comments, so there is no overlap with any tweet corpus by construction. The Model 1 was trained on 2018 HatEval tweets. Davidson (collected 2017) and HateDay (2022) come from other collections. This is independent by origin.

3.5 Evaluation

The evaluation follows the steps of the notebook (Step 4). The two groups have very different label mixes (Table 1), so the first step is to show why overall error rates mislead, and then compare the groups using the tweets labelled *neither*, which are the true negatives for both models.

Main measures. For each group it is computed the false positive rate (FPR) that is the share of tweets that the model flags. The *gap* is the FPR of the AAE-aligned group minus the FPR of the mainstream group. A 95% confidence interval for the gap comes from resampling the tweets 2,000 times (bootstrap). Groups with fewer than 30 tweets are not evaluated.

Steps.

[2] `https://huggingface.co/cardiffnlp/twitter-roberta-base-hate`
[3] `https://huggingface.co/unitary/unbiased-toxic-roberta`

- *Step 4.1.1, baseline evaluation.* Precision, recall, F1 and FPR of each group, with the bootstrap gap.
- *Step 4.1.2, flag rate within each label.* The share of tweets flagged inside each human label (neither, offensive, hate speech), and the gap after matching the label mix of the two groups.
- *Step 4.1.3, comparison without the 0.5 threshold.* The $P(\text{A}>\text{M})$ is the chance that a random AAE-aligned tweet gets a higher score than a random mainstream tweet (0.5 means no difference). It also give the FPR ratio, that measures how many times higher the AAE-aligned FPR is when the mainstream FPR is fixed at 10%.
- *Step 4.1.4, lexical controls.* A keyword control (women-directed profanity) and a vocabulary control (a bag-of-words model that predicts the flag from the words of the tweet).
- *Step 4.1.5, other group definitions.* The same comparison with other ways of defining the groups.
- *Steps 4.2 and 4.3, other corpora.* The same checks on HateXplain and HateDay. For HateDay it was added the controls for profanity and tweet length.
- *Step 4.4, summary.* One table with the main result for the three corpora and the two models.

The confidence intervals only reflect sampling noise, not mistakes of the dialect model or of the human labels.

4 Results

4.1 Baseline evaluation *(Step 4.1.1)*

Table 1 shows why a pooled comparison is misleading. 94.1% of the AAE-aligned tweets are labelled offensive and only 2.0% are labelled neither. In the mainstream group the shares are 64.3% and 28.2%. For model 1, F1 is the same in both groups (0.058 vs. 0.056, gap +0.002, CI $[-0.012, +0.015]$), because the precision is about 0.03 in both. The FPR over all not-hate tweets is very different, 0.751 vs. 0.338 (gap +0.412, CI $[+0.397, +0.428]$). F1 hides the difference, which is why the gap, and not the absolute error, is the finding. For model 2, the positive class is hate or offensive, so its only false positives are the tweets labelled neither. Its FPR is 0.253 vs. 0.161 (gap +0.091, CI $[+0.004, +0.180]$). Its F1 is again equal in both groups (0.885 vs. 0.892, gap −0.006, CI $[-0.015, +0.002]$).

	Label mix (%)		**Model 1 flags (%)**		**Model 2 flags (%)**	
Davidson label	AAE	Main.	AAE	Main.	AAE	Main.
Neither (FPR)	2.0	28.2	9.5	1.6	25.3	16.1
Offensive	94.1	64.3	76.5	48.0	79.7	85.5
Hate speech (recall)	3.9	7.5	58.8	14.9	84.1	85.7

Table 1. Davidson: label mix of the groups and share of tweets flagged by each model within each label.

4.2 Flag rate within each label *(Step 4.1.2)*

Table 1 also gives the share of tweets flagged inside each human label. This removes the effect of the different label mixes. These are descriptive rates, without confidence intervals. The intervals come in Sect. 4.5. Model 1 flags the AAE-aligned group more often in every label. 9.5% vs. 1.6% of the neither tweets, 76.5% vs. 48.0% of the offensive tweets, and 58.8% vs. 14.9% of the hate speech tweets. The gap is therefore not only a matter of label mix. Model 2 is different. It flags the AAE-aligned group more often only for the neither tweets (25.3% vs. 16.1%). For offensive tweets (79.7% vs. 85.5%) and hate tweets (84.1% vs. 85.7%), which are correct detections for a toxicity model, it flags the AAE-aligned group slightly less or equally often. The extra flags of model 2 are therefore on tweets that humans considered harmless.

4.3 Comparison without the threshold *(Step 4.1.3)*

Among the tweets that are not in the model's positive class (*neither* and *offensive* for model 1, *neither* for model 2), the $P(\text{A}>\text{M})$ is 0.761 (CI $[0.753, 0.768]$) for model 1 and 0.656 ($[0.611, 0.707]$) for model 2. With the threshold set so that the mainstream group has an FPR of 10%, the AAE-aligned group has an FPR 3.0 (model 1) and 1.8 (model 2) times as high. The effect is therefore not a product of the default threshold.

4.4 Lexical controls *(Step 4.1.4)*

Model 1 was trained on a corpus that includes hate against women, so it may have learned to flag women-directed profanity (b*tch, h*es), which is frequent in AAE-aligned tweets (91% of the AAE-aligned offensive tweets contain one of 17 such terms vs. 77%).

Keyword control. Among offensive tweets the gap of model 1 is +0.225 with a term and +0.054 without (CIs $[+0.209, +0.241]$ and $[+0.023, +0.085]$). The terms matter a lot but do not account for all of it. Among *neither* tweets, adjusting for the share with a term leaves the gap unchanged ($+0.078 \rightarrow +0.076$), although almost none of those tweets contain a term.

Vocabulary control. A cross-validated bag-of-words logistic regression predicts each classifier's flag from the words of the tweet, on the tweets that are not labelled hate speech. For model 1 it reproduces 89% of the gap (residual +0.046, CI $[+0.035, +0.056]$; +0.036 to +0.087 across tokenizers and regularisation). A ridge regression on the raw score leaves +0.039 (CI $[+0.032, +0.041]$). For model 2 it reproduces 97% (residual +0.004, CI $[-0.002, +0.011]$) because this subset includes the offensive tweets, this is model 2's flag gap on non-hate tweets rather than its false-positive gap. The words most often flagged are profanity, AAE markers (*wit*, *aint*, *tryna*) and, for model 1, gendered words (*she*, *her*). One spelling of the n-word is flagged in only 8.6% of 116 tweets by model 1 while another is flagged 77% of the time (33.6% vs. 97.7% for model 2), so the models react to the form of the word. The groups are defined by words, so this control cannot tell dialect bias from lexical bias.

4.5 Gap within each label and other group definitions *(Step 4.1.5)*

The groups have a different label mix, so this project compares them inside each label. For the model 1, the gap is +0.285 on offensive tweets (CI [+0.269, +0.302]) and +0.078 on *neither* tweets (CI [+0.024, +0.141]). When the mainstream group is given the same label mix, the gap falls from +0.412 to +0.281 (CI [+0.265, +0.296]). So about two thirds of the gap is not caused by the label mix.

Table 2 uses other group definitions. With argmax and with profanity removed before dialect scoring, the *neither* gap stays above zero for both models. Stricter thresholds leave too few *neither* tweets, so it is used the re-weighted gap instead. For model 1 it grows to +0.373 (0.7/0.3) and +0.501 (0.8/0.2). Removing profanity lowers it to +0.166, since many AAE-aligned tweets are in that group because of profanity.

	Model 1			**Model 2**	
Group definition (Davidson)	*neither* gap	std. gap	P(A>M)	*neither* gap	P(A>M)
Baseline 0.6 / 0.4	+.078 [.024,.141]	+.281	.674	+.091 [.004,.180]	.656
Argmax (AAE vs. White)	+.031 [.015,.049]	+.310	.617	+.173 [.135,.211]	.656
Profanity removed before scoring	+.084 [.028,.142]	+.166	.682	+.097 [.005,.188]	.657
Stricter 0.7 / 0.3	–	+.373	–	–	–
Stricter 0.8 / 0.2	–	+.501	–	–	–

Table 2. gap on *neither* tweets (95% CI), label-standardised gap on all not-hate tweets, and P(A>M) on *neither* tweets.

4.6 HateXplain and HateDay *(Steps 4.2 to 4.4)*

Table 3 and Fig. 2 give the same quantities for the three corpora, on the *neither* tweets for both models. In all 12 rows the CI of the gap is above zero and P(A>M) lies between 0.62 and 0.75. The absolute gaps differ by an order of magnitude (HateDay +0.01 to +0.07, HateXplain +0.17 to +0.31) because the base rate of flags differs between corpora. The P(A>M) and the FPR ratios (1.6 to 3.5 when the mainstream FPR is 10%) are comparable across them.

Corpus, groups	Model	n AAE / main.	FPR AAE / main.	Gap [95% CI]	P(A>M)	flagged AAE
Davidson, baseline	1	95 / 3,724	0.095 / 0.016	+0.078 [0.025, 0.142]	0.674	9
	2	95 / 3,724	0.253 / 0.161	+0.091 [0.002, 0.180]	0.656	24
Davidson, argmax	1	626 / 2,688	0.046 / 0.015	+0.031 [0.015, 0.049]	0.617	29
	2	626 / 2,688	0.331 / 0.158	+0.173 [0.133, 0.210]	0.656	207
HateXplain, baseline	1	263 / 4,950	0.369 / 0.121	+0.248 [0.188, 0.308]	0.749	97
	2	263 / 4,950	0.764 / 0.460	+0.304 [0.249, 0.357]	0.700	201
HateXplain, argmax	1	1,047 / 3,744	0.288 / 0.114	+0.174 [0.146, 0.202]	0.657	302
	2	1,047 / 3,744	0.712 / 0.405	+0.306 [0.274, 0.337]	0.705	745
HateDay, baseline	1	286 / 23,652	0.031 / 0.008	+0.024 [0.006, 0.045]	0.733	9
	2	286 / 23,652	0.098 / 0.033	+0.065 [0.031, 0.103]	0.668	28
HateDay, argmax	1	2,436 / 16,075	0.020 / 0.008	+0.012 [0.006, 0.017]	0.667	48
	2	2,436 / 16,075	0.061 / 0.037	+0.025 [0.015, 0.035]	0.641	149

Table 3. Tweets labelled *neither*: FPR in the AAE-aligned and mainstream groups, gap with bootstrap CI, P(A>M) and number of AAE tweets flagged.

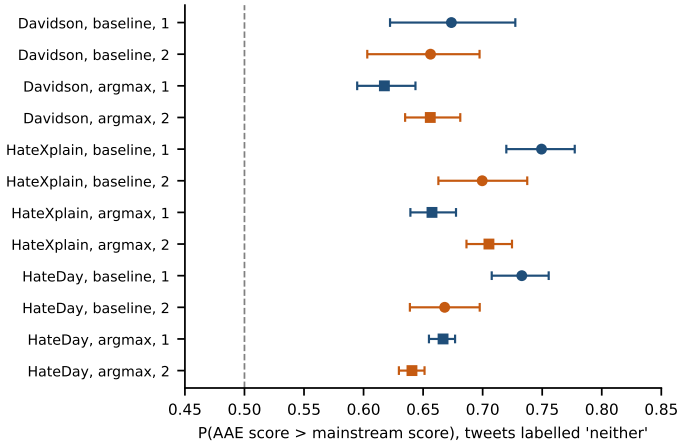


Fig. 2. P(A>M) among *neither* tweets with 95% CI (circles: baseline groups; squares: argmax; blue: model 1; orange: model 2). All intervals lie above 0.5.

HateDay in detail. HateDay is a random sample, and the results differ from Davidson. The AAE-aligned group is small (397 tweets, 1.6% of the sample) and has only 2 hate tweets, so recall cannot be tested. Profanity is rare in both groups (4.9% vs. 0.6% of *neither* tweets), and AAE-aligned tweets are shorter (median 4 vs. 10 words). Even so, the gap remains on tweets *without* profanity (+0.022 [+0.003, +0.043] for model 1, +0.033 [+0.006, +0.061] for model 2) and after matching on length and profanity (+0.023 and +0.040). It is smallest in very short tweets and grows with length. Unlike on Davidson, the vocabulary explains only 12% and 21% of the flag gap (44% and 37% of the score gap). The words most tied to the AAE-aligned group are harmless slang (*fr, gm, nah, fam*) that is rarely flagged. The model 1 gap rests on only 9 flagged AAE-aligned tweets, so P(A>M) is the more reliable number.

5 Discussion

5.1 Answers to the research questions

1. **Are AAE-aligned tweets that are not hate flagged more often?** Yes. On Davidson, model 1 flags 9.5% of the AAE-aligned *neither* tweets and 1.6% of the mainstream ones; model 2 flags 25.3% and 16.1%. The AAE-aligned tweets also get higher scores. A random AAE-aligned tweet scores higher than a random mainstream one with probability 0.62 to 0.75. F1 and accuracy hide this, because F1 is the same in both groups.
2. **Does it happen for both classifiers?** Yes. The gap is positive, with a confidence interval above zero, for model 1 (hate detector) and for model

2 (toxicity detector) in every comparison. It is therefore not specific to one model or to one definition of harmful content.

3. **Does it happen in all three corpora?** Yes, in all 12 comparisons (3 corpora, 2 models, 2 group definitions). The size of the gap differs a lot, from +0.01 to +0.07 on HateDay and from +0.17 to +0.31 on HateXplain. This is because the corpora have very different shares of flagged tweets. P(A>M), which does not depend on that share, is similar in all three (0.62 to 0.75).
4. **Does it stay with a different threshold?** Yes. P(A>M) uses no threshold and is above 0.5 in all rows. When the threshold is set so that the mainstream group has an FPR of 10%, the AAE-aligned FPR is 1.6 to 3.5 times higher. The gap is not caused by the default threshold of 0.5.
5. **Does it stay with other group definitions?** Yes. With argmax and with profanity removed before dialect scoring, the gap on *neither* tweets stays above zero for both models. With stricter thresholds there are too few AAE-aligned *neither* tweets, but the re-weighted gap on all not-hate tweets grows (model 1: +0.373 and +0.501). The more clearly a tweet matches AAE, the larger the gap.
6. **Is it caused by the label mix?** Only in part. On Davidson, 94% of the AAE-aligned tweets are offensive, against 64% of the mainstream ones. When the mainstream group is given the same label mix, the model 1 gap falls from +0.412 to +0.281. About a third of the gap comes from the label mix, and two thirds remain. Inside each label the gap is still there (+0.285 on offensive tweets and +0.078 on *neither* tweets).
7. **Is it caused by profanity?** In part on Davidson, not on HateDay. On Davidson, profanity aimed at women matters a lot, among offensive tweets, the model 1 gap is +0.225 for tweets with such a word and +0.054 for tweets without. Removing profanity before dialect scoring lowers the re-weighted gap to +0.166. On HateDay, profanity is rare, and the gap remains on tweets without any profanity (+0.022 for model 1, +0.033 for model 2).
8. **Is it caused by tweet length?** No. This was tested on HateDay, where AAE-aligned tweets are much shorter (median 4 vs. 10 words). After matching the groups on length and profanity, the gap remains (+0.023 for model 1, +0.040 for model 2). The gap is smallest in very short tweets and largest in long ones, so short length does not create the gap.
9. **Do the words explain the gap?** On Davidson, mostly yes, a bag-of-words model reproduces 89% (model 1) and 97% (model 2) of the flag gap. On HateDay, mostly no, it reproduces only 12% and 21% of the flag gap, and 44% and 37% of the score gap. However, the groups are themselves defined by words, so this control cannot separate dialect bias from word bias.
10. **Which words go with the false positives?** On Davidson, profanity, words aimed at women, AAE words such as *wit*, *aint* and *tryna*, and for model 1 also *she* and *her*. The spelling matters, one spelling of the n-word is flagged by model 1 in 8.6% of tweets, another in 77%. On HateDay, the words most typical of the AAE-aligned group are harmless slang (*fr*, *gm*, *nah*, *fam*), and they are rarely flagged.

5.2 Further discussion

The gap is robust, it appears for both models, in all corpora, at any threshold and with any group definition. What causes it differs between corpora. On Davidson, a lexicon-selected corpus, the flags mostly follow profanity and AAE words. On HateDay, a random sample, words, profanity and length explain little of it. This suggests that the models react to more than single words, but with only 9 flagged AAE-aligned tweets for model 1 the evidence is thin. Model 1 detects hate against immigrants and women, model 2 detects toxicity, and Davidson separates hate from offensive language. These differences can create a gap on their own. For example, model 1 probably learned to flag profanity aimed at women, which is more common in AAE-aligned tweets. This is why the main results use the *neither* tweets, where no definition of hate applies. I read the gap as a property of the systems as they are used, not as a finding about hate speech itself.

Limitations.

1. The project only audits false positives. Recall cannot be tested on HateDay and is unreliable on the other corpora.
2. The dialect score measures how similar the words are to AAE, not who wrote the tweet. Because the groups are defined by words, the word-based control cannot separate dialect bias from word bias. The baseline mainstream group also contains non-English tweets.
3. The human labels are noisy. HateDay has one annotator per tweet, and the Davidson AAE-aligned *neither* group is small (95 tweets).
4. Toxicity is not hate, so the results for model 2 are about toxicity detection.
5. Duplicates and retweets were not verified.
6. The confidence intervals only cover sampling noise. They ignore errors of the dialect model and of the labels.

A causal test would need the same content written in both AAE and mainstream English. No such data exists for these corpora.

6 The Final Report Compared with the Proposal

The core of the proposal is done and extended. I compare AAE-aligned and mainstream tweets with a pretrained classifier, labelled tweets and a dialect tool. I use two classifiers instead of one, three corpora instead of one, bootstrap confidence intervals and measures that do not depend on the threshold. The dialect tool of Blodgett et al. was ported to Python 3 and, as planned, its score is treated as word similarity, not as the identity of the author. The search for words tied to false positives is done. Masking words and writing pairs of AAE and mainstream tweets were not done, because they need new data. The expected outcome, that the classifier does worse on AAE tweets, is confirmed for false positives. It is not visible in F1, and it cannot be tested for recall.

The proposal planned to use the Waseem and Hovy dataset. It is shared only as tweet IDs, and downloading old tweets from the X API is no longer free

(about $95, and many tweets are gone). Davidson includes the text, so it became the main corpus. HateXplain and HateDay were added to test if the result holds on other data. HateDay is a random sample, so it balances the lexicon-selected Davidson sample. The proposal also planned to compare how well the model does on each group. Accuracy and F1 are not fair here, because the classifiers definitions of hate differ from the labels (Table 1). The FPR gap became the main measure, as agreed with Rahil.

Fine-tuning on a balanced sample was moved to future work, as agreed with Rahil. Fine-tuning on Davidson would change the classifier's definition of hate to Davidson's, so I couldn't tell a drop in bias from a change in what counts as hate. This needs its own design. The time I had, on CPU only, went into the second classifier and the random sample instead.

Rahil feedback

- *Name the classifiers, their training data, and the label mapping.* Done in Sect. 3. A first version of model 2 had been trained on Davidson's tweets, so it was replaced.
- *Make sure the data is available.* Waseem and Hovy was replaced by Davidson. HateDay is gated, and none of its tweets is stored in the repository.
- *Narrow the scope.* The work focuses on one question, the false positive gap, and on what explains it. Fine-tuning was dropped. I use one family of measures, the FPR gap and $P(\text{A}>\text{M})$.
- *Use a second model.* Model 2 was added. The gap appears for both models (Sect. 4.6).
- *State the definition mismatch (profanity aimed at women).* Discussed above and measured with the keyword control (Sect. 4.4).
- *Compare the FPR within each label.* Done in Sect. 4.2 and Sect. 4.5. The gap remains inside each label, and about a third of the overall gap comes from the label mix.

Future work.

- Fine-tune on a sample balanced by dialect and class, and compare it with changing the threshold per group, while keeping the definition of hate fixed.
- Mask the flagged words and run the classifiers again, and write pairs of the same tweet in AAE and mainstream English.
- Remove duplicates and retweets, check overlap with the training data at the text level, and read the flagged HateDay tweets by hand (without sharing them).

7 Conclusion

Two pretrained classifiers, one for hate and one for toxicity, flag AAE-aligned tweets that humans labelled *neither* more often than mainstream tweets. This happens in three corpora and in all 12 comparisons. The label mix, the threshold and the way the groups are defined do not remove the gap. On Davidson, the

flags mostly follow the words in the tweet. On the random sample, the gap is not explained by the words, but it rests on few flagged tweets. I cannot say whether the classifiers react to the dialect itself, and I leave mitigation to future work.

References

[1] Su Lin Blodgett, Lisa Green, and Brendan O'Connor. Demographic dialectal variation in social media: A case study of African-American English. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1119–1130, 2016.

[2] Thomas Davidson, Dana Warmsley, Michael Macy, and Ingmar Weber. Automated hate speech detection and the problem of offensive language. In *Proceedings of the 11th International AAAI Conference on Web and Social Media (ICWSM)*, pages 512–515, 2017.

[3] Binny Mathew, Punyajoy Saha, Seid Muhie Yimam, Chris Biemann, Pawan Goyal, and Animesh Mukherjee. HateXplain: A benchmark dataset for explainable hate speech detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 14867–14875, 2021.

[4] Maarten Sap, Dallas Card, Saadia Gabriel, Yejin Choi, and Noah A. Smith. The risk of racial bias in hate speech detection. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 1668–1678, 2019.

[5] Manuel Tonneau, Diyi Liu, Niyati Malhotra, Scott A. Hale, Samuel P. Fraiberger, Victor Orozco-Olvera, and Paul Röttger. HateDay: Insights from a global hate speech dataset representative of a day on Twitter. arXiv preprint arXiv:2411.15462, 2024.